SAMSUNG

Together for Tomorrow! Enabling People
Education for Future Generations

Credit Risk Prediction: A Loan Default Classification Model

SIC – AI 801 & 802

Agenda

- Team Members
- Problem & Dataset
- Data cleaning & Pre-processing
- Exploratory Data Analysis (EDA)
- Machine Learning Model & Evaluation
- Business Recommendations
- Limitations
- Future Work & Enhancement
- Conclusion

Team Members

Samah Mahmoud gaber

Toka Abdelaziz Mohamed

Esraa Mostafa El Tohamy

Ammar Yasser Khaled

Problem & Dataset

Problem & Dataset

Problem

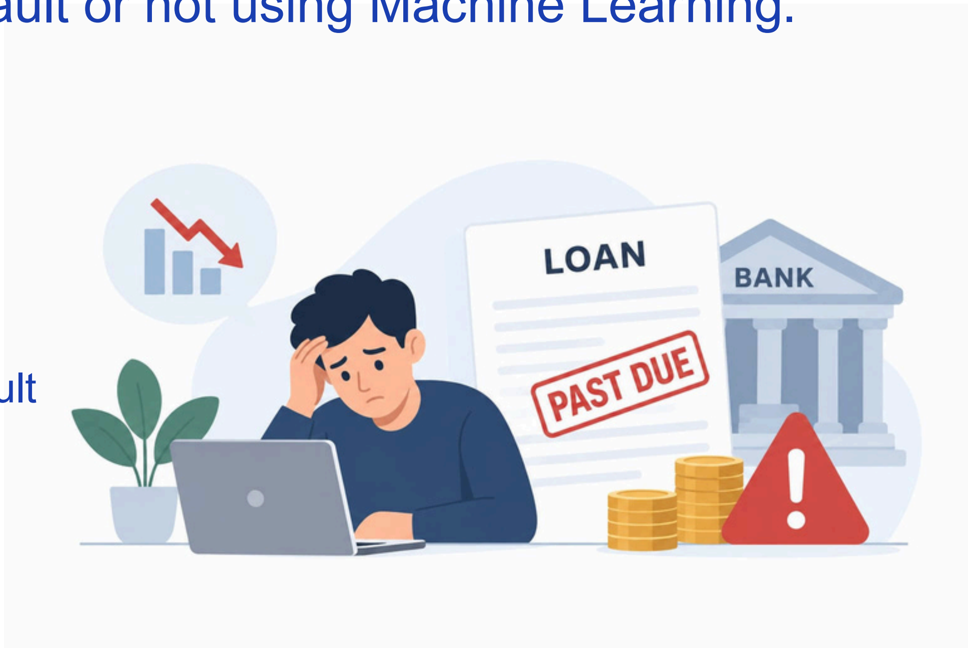
Financial institutions face the risk of customers defaulting on their loans.

Goal:

Predict whether a customer will default or not using Machine Learning.

Dataset:

- Name: Credit Risk Prediction
- Rows: 32,581
- Columns: 12
- Target: loan_status → Default / No Default
- Contains information about
- the customer and the loan.



Data cleaning & Pre-processing

Data cleaning

Data Quality Issues Identified

- **Missing Values:**

  Missing values were found in person_emp_length (895) and loan_int_rate (3116).

  → Kept the observations and handled missing numerical values later using median imputation.

- **Duplicate Records:**

  165 exact duplicate rows were identified.

  → Removed to prevent repeated observations from biasing the model.

- **Logically Invalid Values:**

  5 observations had age > 100 years.

  2 observations had 123 years of employment, which was inconsistent with their age.

  9 observations had a zero loan-to-income ratio despite having positive income and loan amounts.

  → These observations were dropped

```
Age < 18: 0
Age > 100: 5
Income <= 0: 0
Employment Length < 0: 0
Employment Length > 50: 2
Loan Amount <= 0: 0
Interest Rate <= 0: 0
Loan % Income <= 0: 9
Credit History Length < 0: 0
```

Outlier Handling & Feature Engineering

Extreme Income Values

- person_income has a strong right-skewed distribution
- The maximum income reached 6,000,000, while the median was approximately 55,000.

Approach

- The 99.5th percentile of the training data was used as the extreme-value threshold.
- The affected income values were treated as missing and later handled through median imputation.

Why?

- Deleting the entire observation could remove useful information.
- The threshold was calculated using training data only to prevent data leakage.

Feature Engineering

Created Employment-to-Age Ratio:

- Represents employment experience relative to the borrower's age.
- Provides more context than employment length alone.

Data Transformation & Preprocessing Pipeline

Numerical Features

- Missing numerical values were handled using Median Imputation because the numerical variables contain skewed distributions and median is less sensitive to extreme values.
- For scale-sensitive models, numerical features were standardized using StandardScaler.

Categorical Features

- One-Hot Encoding was applied to nominal categorical variables:

Ordinal Feature

- loan_grade was encoded using an ordinal mapping (A → G)

Model-Specific Preprocessing

- **Scaled preprocessing Pipline:** Logistic Regression because it is sensitive to feature scale.
- **Unscaled preprocessing Pipline:** Tree-based models such as Decision Tree and Random Forest because they do not require feature scaling.

Exploratory Data Analysis (EDA)

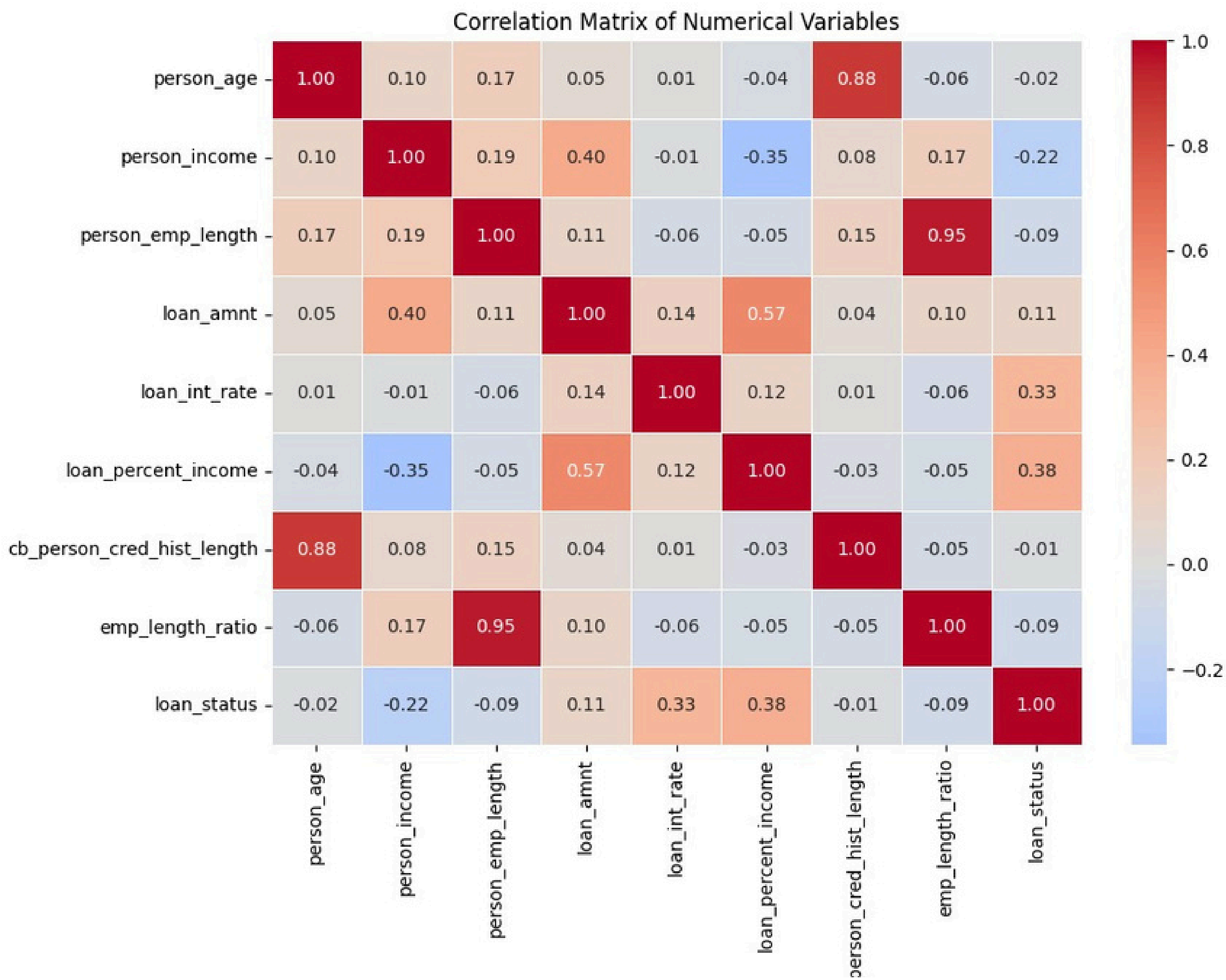
Descriptive Statistics

Key Insight:

- Four numerical variables show right-skewed distributions: Person Income, Loan Amount, Loan Percent Income, and Employment Length Ratio.

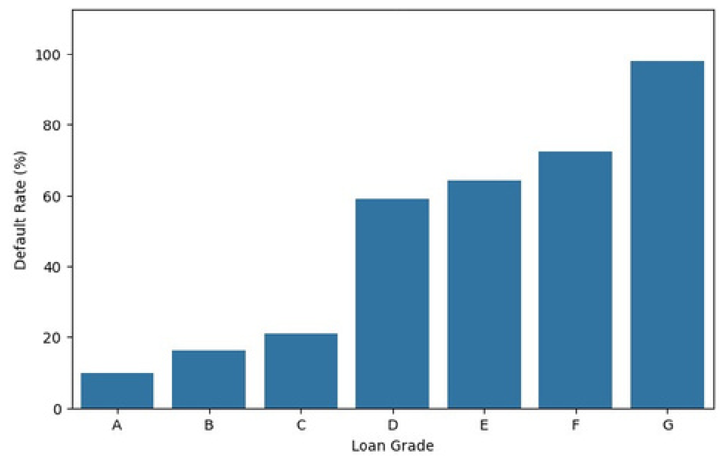
	count	mean	median	std	min	25%	50%	75%	max
person_age	25927.0	27.71	26.00	6.18	20.00	23.00	26.00	30.00	94.00
person_income	25802.0	63487.48	55000.00	36679.82	4000.00	38400.00	55000.00	78000.00	294000.00
person_emp_length	25203.0	4.76	4.00	4.01	0.00	2.00	4.00	7.00	38.00
loan_amnt	25932.0	9592.17	8000.00	6286.93	500.00	5000.00	8000.00	12250.00	35000.00
loan_int_rate	23444.0	11.02	10.99	3.23	5.42	7.90	10.99	13.48	23.22
loan_percent_income	25926.0	0.17	0.15	0.11	0.01	0.09	0.15	0.23	0.83
cb_person_cred_hist_length	25932.0	5.80	4.00	4.05	2.00	3.00	4.00	8.00	30.00
emp_length_ratio	25198.0	0.17	0.14	0.14	0.00	0.06	0.14	0.27	0.72

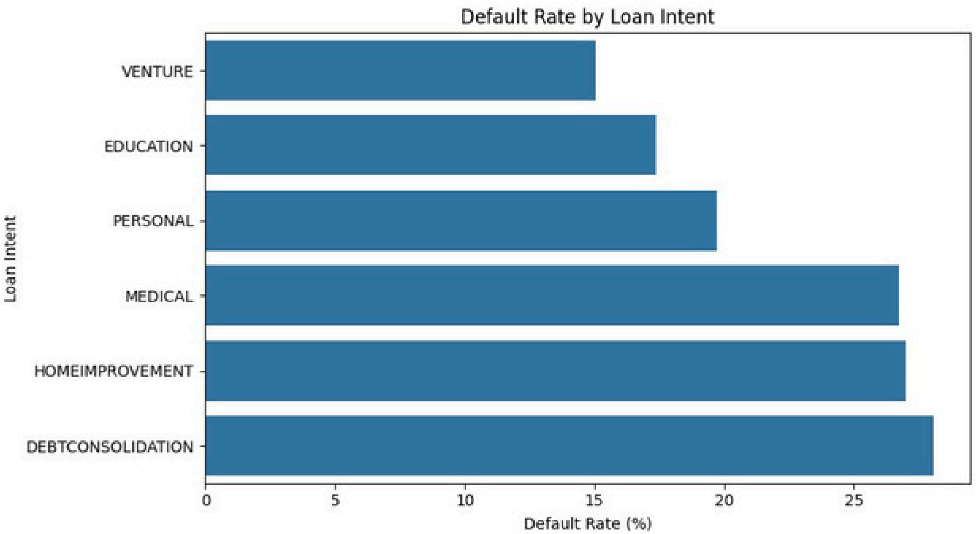
Correlation Analysis

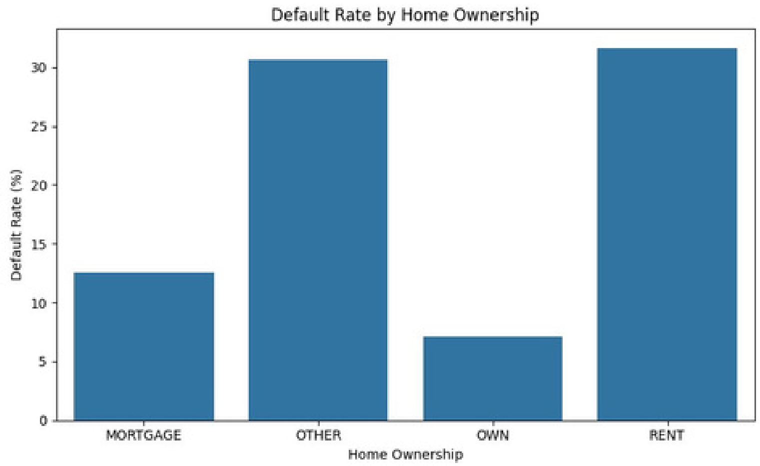


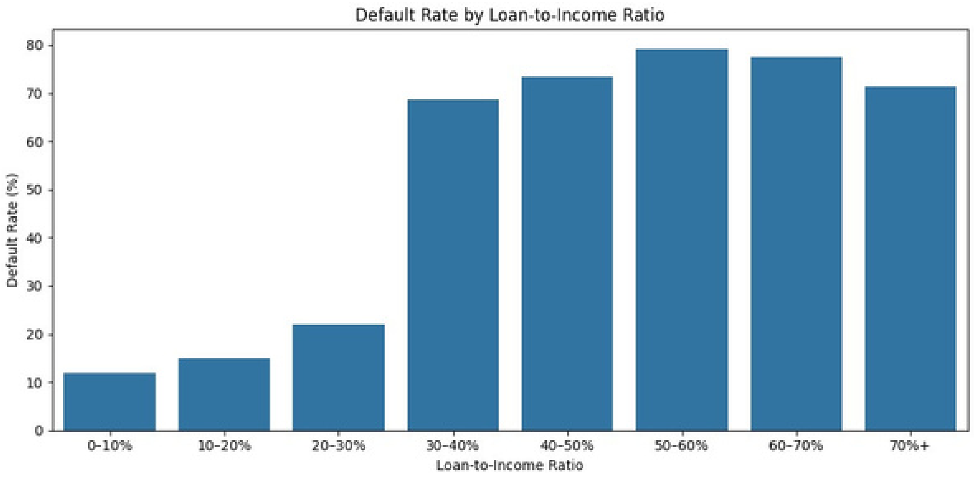
Relationships

Loan Grade → Default ↑

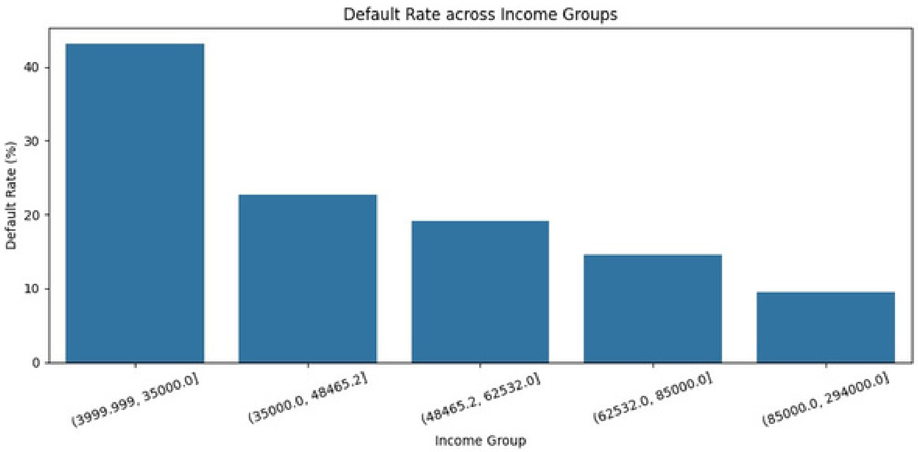


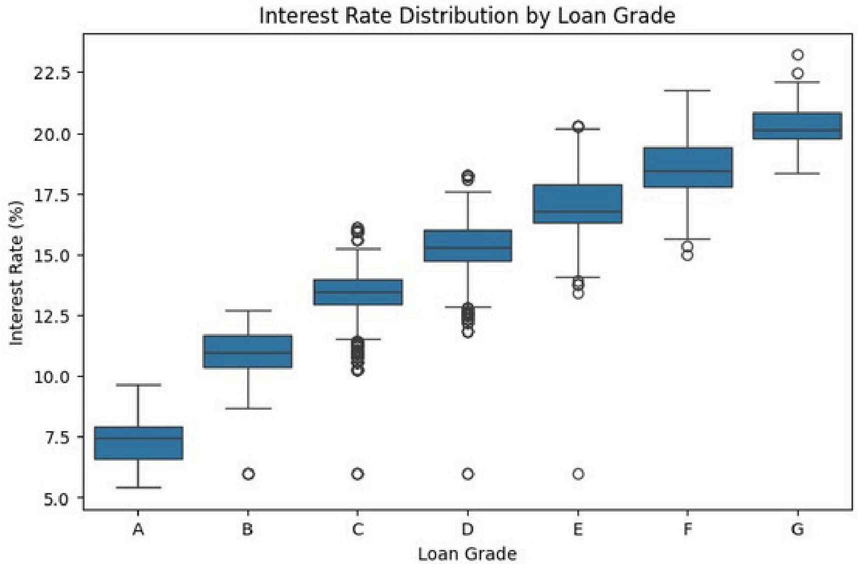


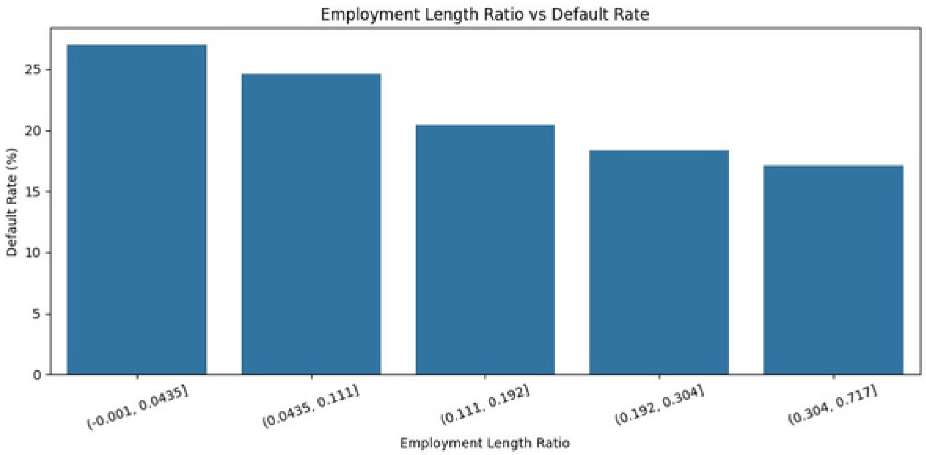




Relationships







Key Insight:

- Income: Higher income → Lower default rate.
- Loan Grade & Interest Rate: Higher-risk grades → Higher interest rates.
- Employment Length Ratio: Higher ratio → Lower default rate

Machine Learning Model & Evaluation

Problem Classification

Customer Data:
1.Age
2.Income
3.Loan Amount

Model:
1.Logestic Regression
2.Random Forest
3.XGBoost

0 : Low Risk
Non - Default

1 : High Risk
Default

<u>Core Objective:</u> Identifying potential loan defaulters to mitigate financial risks for lending institutions.

<u>Problem Type:</u> Binary Classification.

<u>Target Variable:</u> loan_status
Class 0: Non-default (Safe Borrower).
Class 1: Default (High-Risk Borrower)..

<u>The Challenge:</u> Highly imbalanced dataset consisting of ~78% non-defaulters versus ~22% defaulters.

Model Training & Class Imbalance

Default cases represent a smaller portion of the data, making the model biased toward the majority class.

Normal

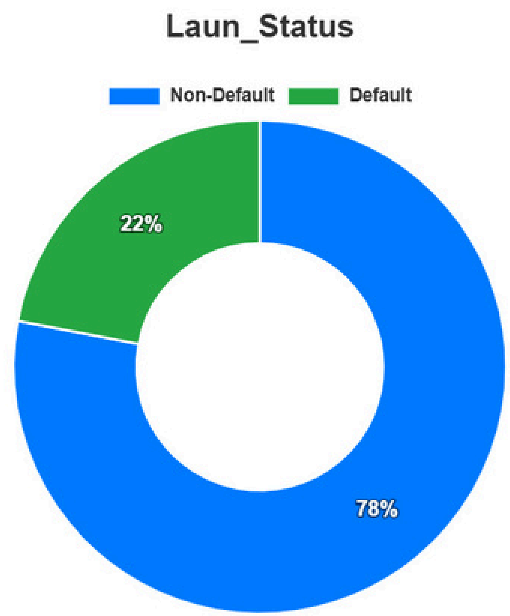
Train using the original class distribution

Weighted

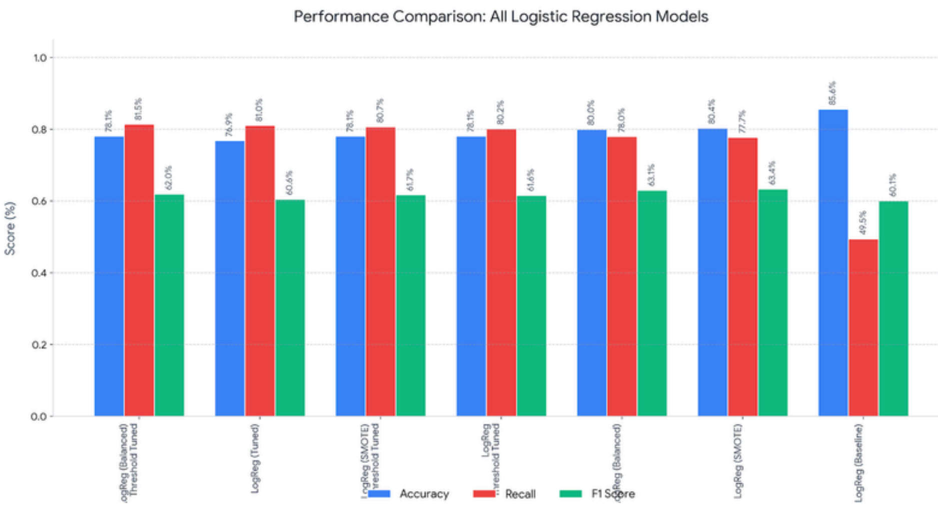
Give higher importance to the minority class

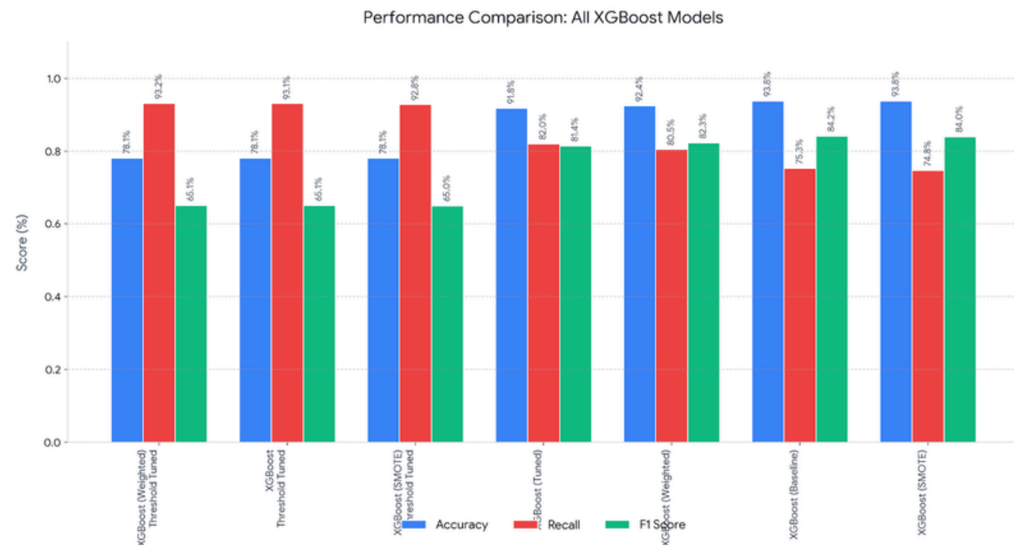
SMOTE

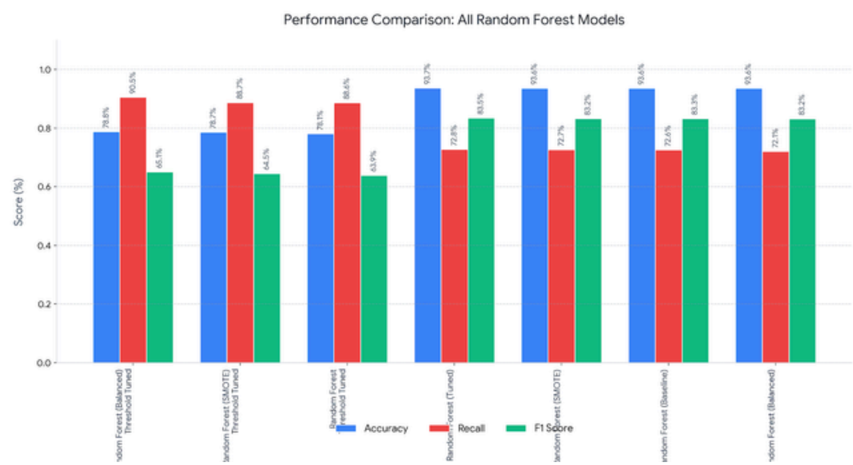
Generate synthetic samples for the minority class



Model Selection & Experimentation

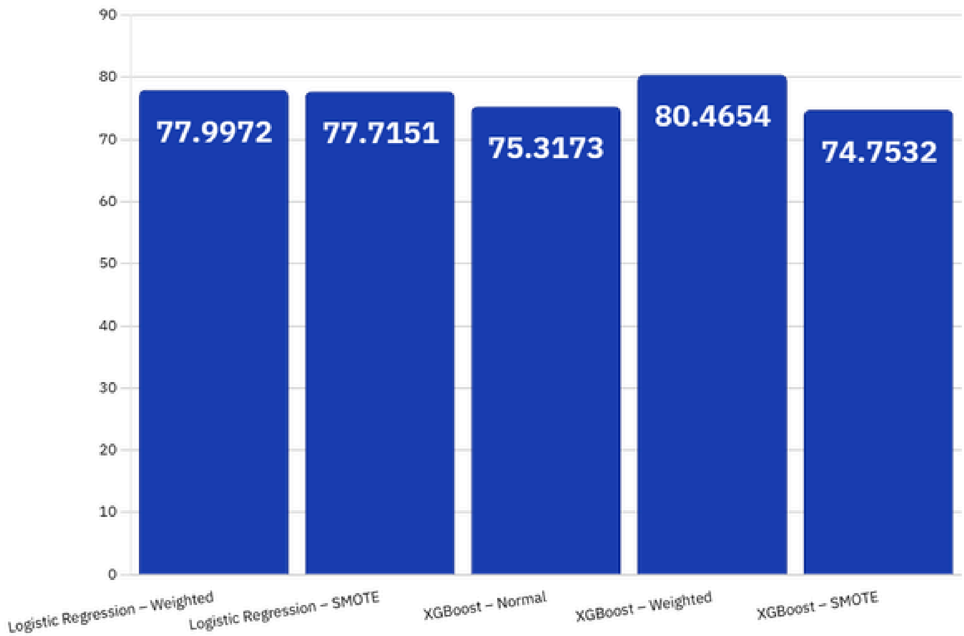




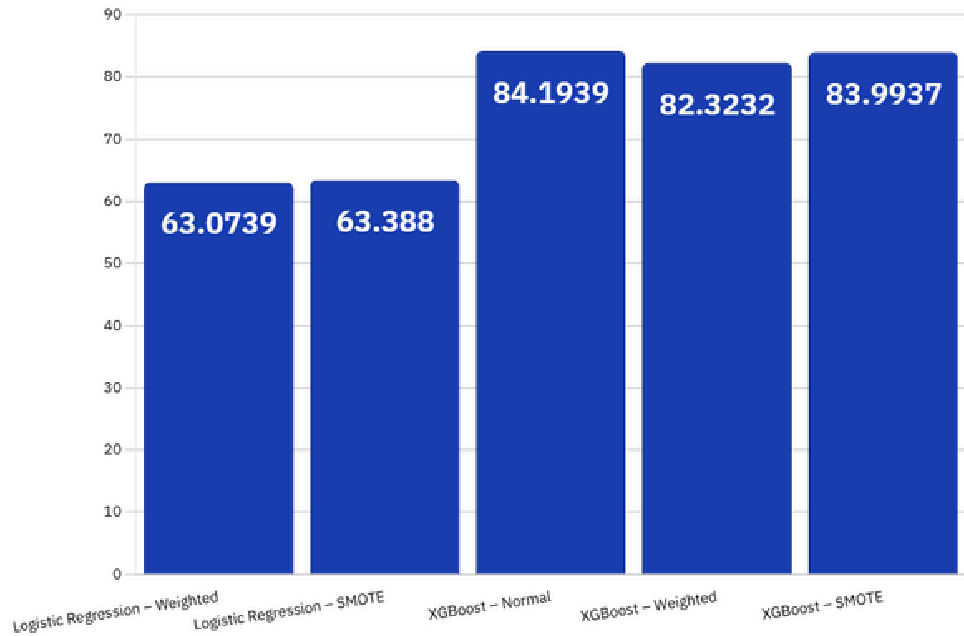


Best Model Selection

Recall Value



F1-Score Value



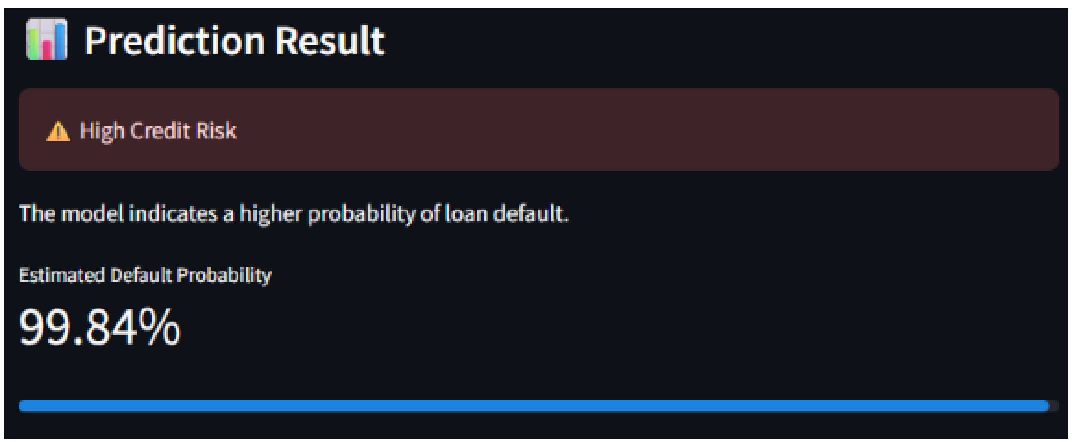
Model Deployment

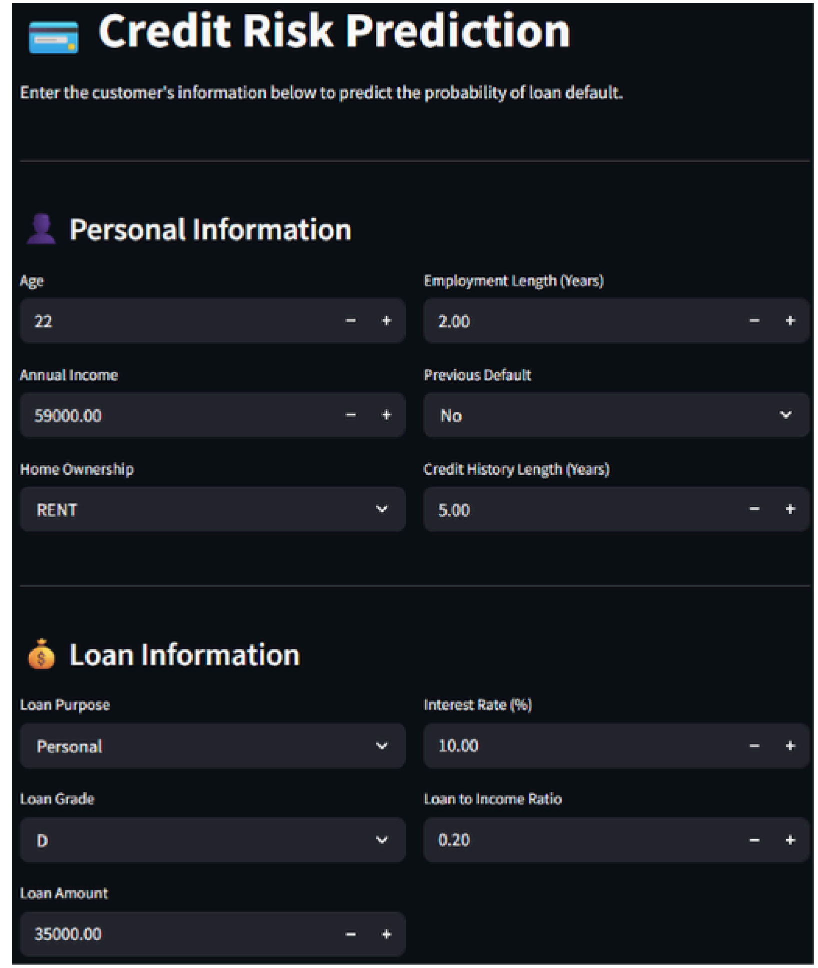
Deployment Challenge

The model was trained on 24 transformed features, while the application initially provided only 12 original features.

Solution

We ensured that the same preprocessing and feature transformation pipeline used during training is applied to user inputs before prediction.





Business Recommendations

Business Recommendations

AI as an Assistant:
Use the Streamlit application to empower loan officers, not replace human judgment.

Dynamic Risk Pricing:
Adjust interest rates based on predicted risk, and strictly flag applicants with a Loan-to-Income ratio > 20%.

Continuous Auditing:
Regularly review False Positives and False Negatives to maintain system reliability and prevent financial loss.

Limitations

Limitations

Feature Scope Constraints:
The dataset relies exclusively on individual borrower metrics, lacking external economic context.

Model Interpretability (Black-Box):
The champion XGBoost model sacrifices simple transparency to achieve its high 92.4% accuracy.

Static Training Data:
The model is trained on a fixed historical snapshot, making it vulnerable to performance degradation over time (Data Drift).

Future Work & Enhancements

Future Work & Enhancements

Macroeconomic Integration:
Incorporate inflation and unemployment rates to build economic resilience against market shifts.

Explainable AI (XAI):
Integrate SHAP values into the GUI to explicitly show why a borrower is flagged.

Automated MLOps via n8n:
Orchestrate an automated workflow using n8n to fetch new data, retrain the model, and prevent data drift.

Conclusion

Conslusion

Objective Achieved:
Built a robust credit risk classifier handling ~32K imbalanced records.

Key Risk Drivers:
Loan-to-Income Ratio, Interest Rate, and Loan Grade.

Best Fit Model:
XGBoost (Weighted), selected from 21 experiments.

Top Performance:
Achieved 80.5% Recall and 92.4% Accuracy, effectively minimizing costly false negatives.

Practical Deployment:
Delivered a real-time Streamlit GUI for instant, actionable risk insights

Thank You

SAMSUNG

Together for Tomorrow!
Enabling People

Education for Future Generations